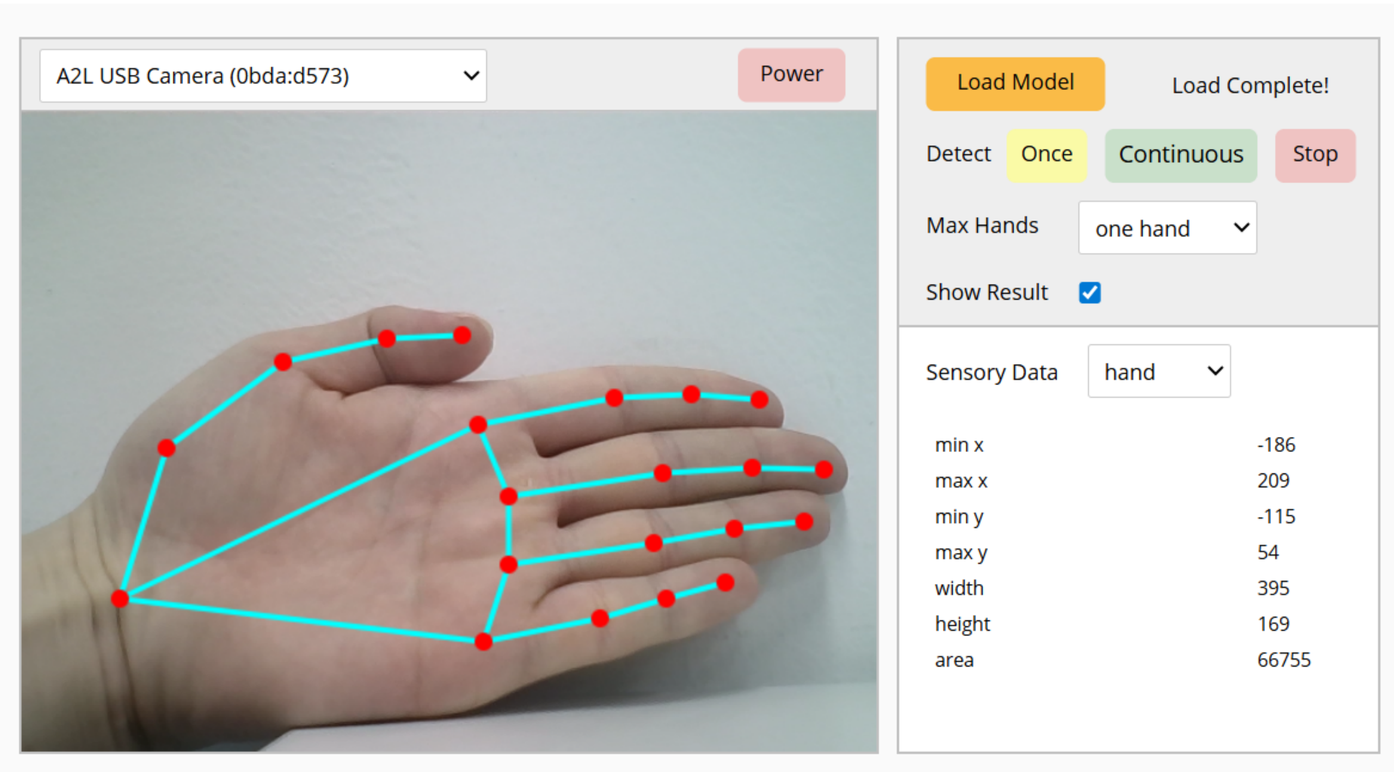
Open Dashboard

Open Dashboard is not a block that can be used in block coding,
but it allows you to open a dashboard where you can check how the model used in the extension module is applied.

Dashboard Screen

When you click Open Dashboard, you can see the following screen.



Detailed Features

Power

Turns the selected camera on or off.

Load Model

Loads the trained hand model. This is a required step to use the **Hand Detection** extension module.

Detect

Starts or stops hand detection.
You can choose whether to run it only once with the Once button, or continuously with the Continuous

button.
You can also stop detection using the Stop button.

Show Result

Displays hand detection results on the camera screen.

Max Hands

Selects the maximum number of hands to detect.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
hand	Dropdown Option	Number of hands to detect	one hand, both hands

Sensory Data

Displays data values based on the selected hand part.

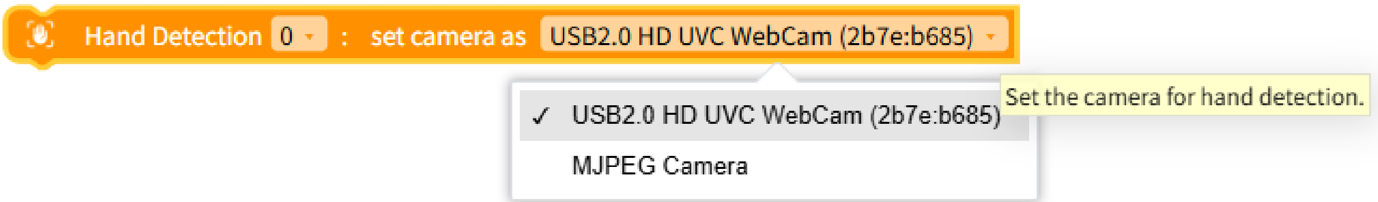
Dropdown Options and Input Values

Name	Type	Description	Range / Type
parts	Dropdown Option	Hand part	hand, palm, wrist, thumbs, index, middle, ring, pinky

Blocks

Select Camera

Selects the camera to use for the Hand Detection module.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
camera	Dropdown Option	Camera to use	Connected camera list

JavaScript Code

```javascript
// Set a specific camera for hand detection (id is an example)
$('HandDetection*0:camera.deviceId').d =
'035658da47183882a695a82c45b8f3e9ae50cef47945ccdc3f31e1ae1fbca9cb';
```

Python Code

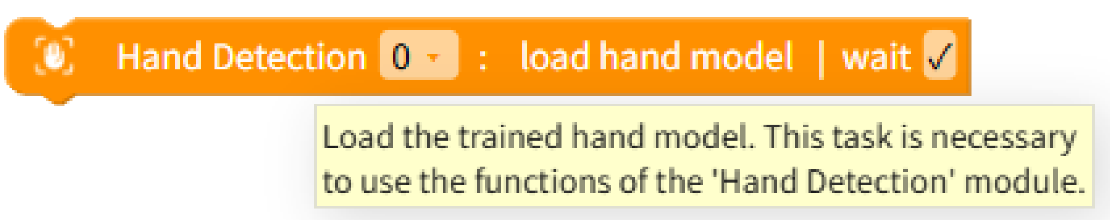
```python
# Set a specific camera for hand detection (id is an example)
__('HandDetection*0:camera.deviceId').d =
'035658da47183882a695a82c45b8f3e9ae50cef47945ccdc3f31e1ae1fbca9cb'
```

Load Hand Model

Loads the trained hand model. This must be done to use the features of the Hand Detection module.

If **Wait** is checked, it waits until the model finishes loading.

If **Wait** is checked, it can only be used inside an async function.



JavaScript Code

```
// Load hand model | Wait: O
$('HandDetection*0:load_model').d = 1;
await $('HandDetection*0:!load_model').w();

// Load hand model | Wait: X
$('HandDetection*0:load_model').d = 1;
```

Python Code

```
# Load hand model | Wait: O
__('HandDetection*0:load_model').d = 1
await __('HandDetection*0:!load_model').w()

# Load hand model | Wait: X
__('HandDetection*0:load_model').d = 1
```

Set Detection Target

Decides whether to detect based on **one hand** or **both hands**.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
pivot	Dropdown Option	Detection target	One hand (1), Both hands (2)

JavaScript Code

```
// Set to detect one hand
$('HandDetection*0:max_count').d = 1;

// Set to detect both hands
$('HandDetection*0:max_count').d = 2;
```

Python Code

```
# Set to detect one hand
__('HandDetection*0:max_count').d = 1

# Set to detect both hands
__('HandDetection*0:max_count').d = 2
```

Detect Hands Once

Detects hands currently on the screen and displays them only once.

JavaScript Code

```
// Detect hands once
$('HandDetection*0:detect.once').d = 1;
```

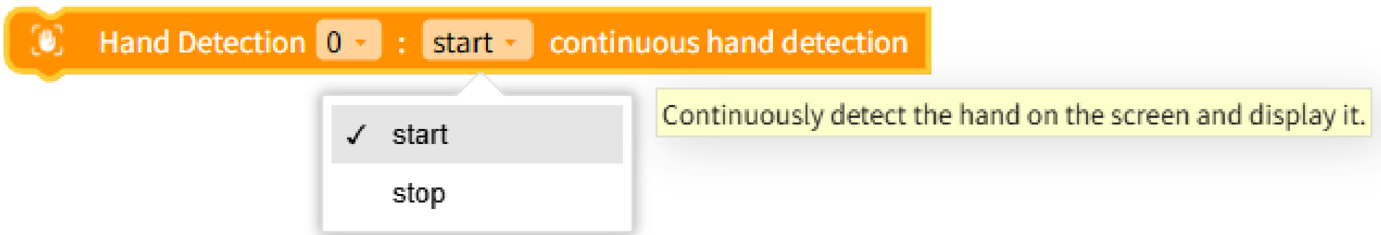
Python Code

```
# Detect hands once
__('HandDetection*0:detect.once').d = 1
```

Detect Hands Continuously

Starts or stops continuous hand detection.

When continuous detection starts, it keeps tracking and displaying hands currently on the screen.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
toggle	Dropdown Option	Hand detection	Start (1), Stop (0)

JavaScript Code

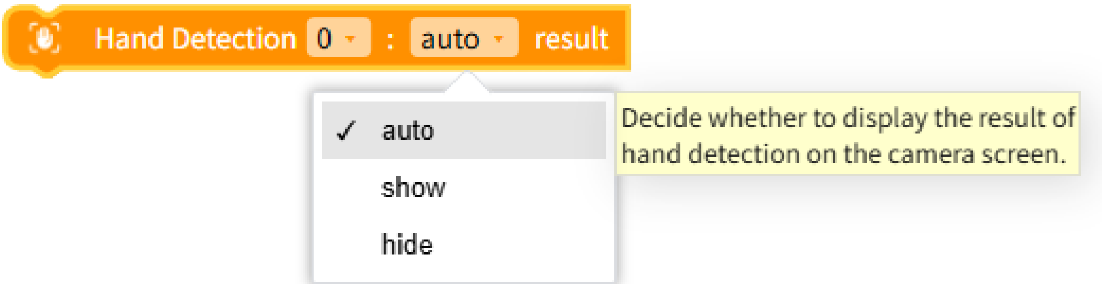
```
// Start continuous hand detection
$('HandDetection*0:detect.continuous').d = 1;

// Stop continuous hand detection
$('HandDetection*0:detect.continuous').d = 0;
```

Python Code

```
# Start continuous hand detection
__('HandDetection*0:detect.continuous').d = 1

# Stop continuous hand detection
__('HandDetection*0:detect.continuous').d = 0
```

Show Hand Detection Result

Decides whether to display hand detection results on the camera screen.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
toggle	Dropdown Option	Show results	Show (1), Hide (0)

JavaScript Code

```
// Show hand detection results
$('HandDetection*0:display').d = 1;

// Hide hand detection results
$('HandDetection*0:display').d = 0;
```

Python Code

```
# Show hand detection results
__('HandDetection*0:display').d = 1

# Hide hand detection results
__('HandDetection*0:display').d = 0
```

Hand-Related Data

Returns the data of the selected hand part.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction	Dropdown Option	Hand side	left, right
part	Dropdown Option	Hand part	palm, wrist
axis	Dropdown Option	Coordinate axis	x, y

JavaScript Code

```
// Right palm x coordinate
$('HandDetection*0:right.palm.x').d;

// Right palm y coordinate
$('HandDetection*0:right.palm.y').d;

// Right wrist x coordinate
$('HandDetection*0:right.wrist.x').d;

// Right wrist y coordinate
$('HandDetection*0:right.wrist.y').d;

// Left palm x coordinate
$('HandDetection*0:left.palm.x').d;

// Left palm y coordinate
$('HandDetection*0:left.palm.y').d;

// Left wrist x coordinate
```

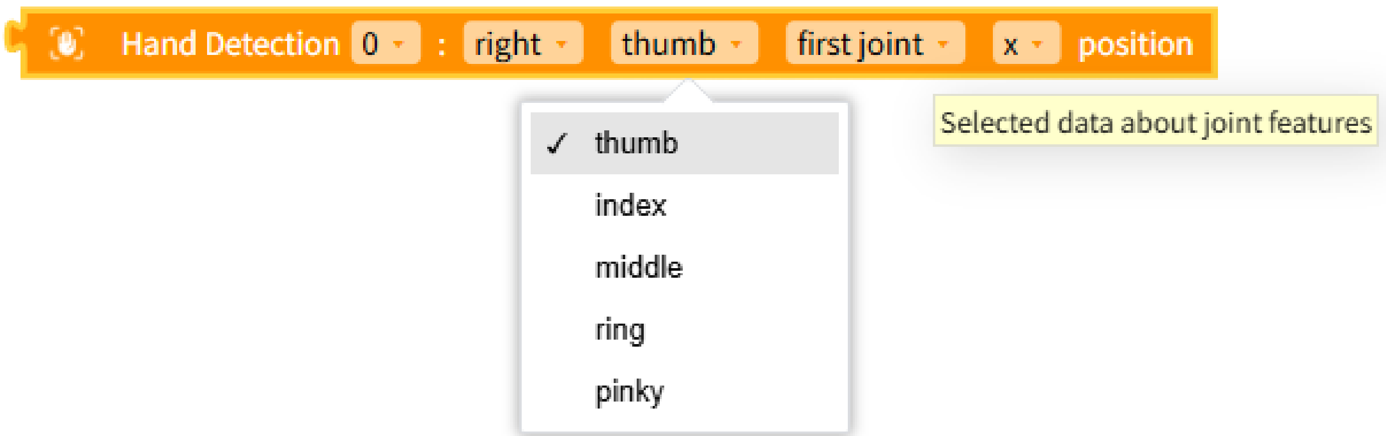
```
$('HandDetection*0:left.wrist.x').d;

// Left wrist y coordinate
$('HandDetection*0:left.wrist.y').d;
```

Python Code

```
# Right palm x coordinate
__('HandDetection*0:right.palm.x').d

# Right palm y coordinate
__('HandDetection*0:right.palm.y').d

# Right wrist x coordinate
__('HandDetection*0:right.wrist.x').d

# Right wrist y coordinate
__('HandDetection*0:right.wrist.y').d

# Left palm x coordinate
__('HandDetection*0:left.palm.x').d

# Left palm y coordinate
__('HandDetection*0:left.palm.y').d

# Left wrist x coordinate
__('HandDetection*0:left.wrist.x').d

# Left wrist y coordinate
__('HandDetection*0:left.wrist.y').d
```

Finger Joint Data

Returns the selected finger joint data.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction	Dropdown Option	Hand side	left, right
finger	Dropdown Option	Finger	thumb, index, middle, ring, pinky
joint	Dropdown Option	Finger joint	first, second, third, last
axis	Dropdown Option	Coordinate axis	x, y

JavaScript Code

```
// Right thumb first joint x coordinate
$('HandDetection*0:right.thumb.first.x').d;

// Right index second joint x coordinate
$('HandDetection*0:right.index.second.x').d;

// Right middle third joint x coordinate
$('HandDetection*0:right.middle.third.x').d;

// Right ring first joint y coordinate
$('HandDetection*0:right.ring.first.y').d;

// Right pinky first joint y coordinate
```

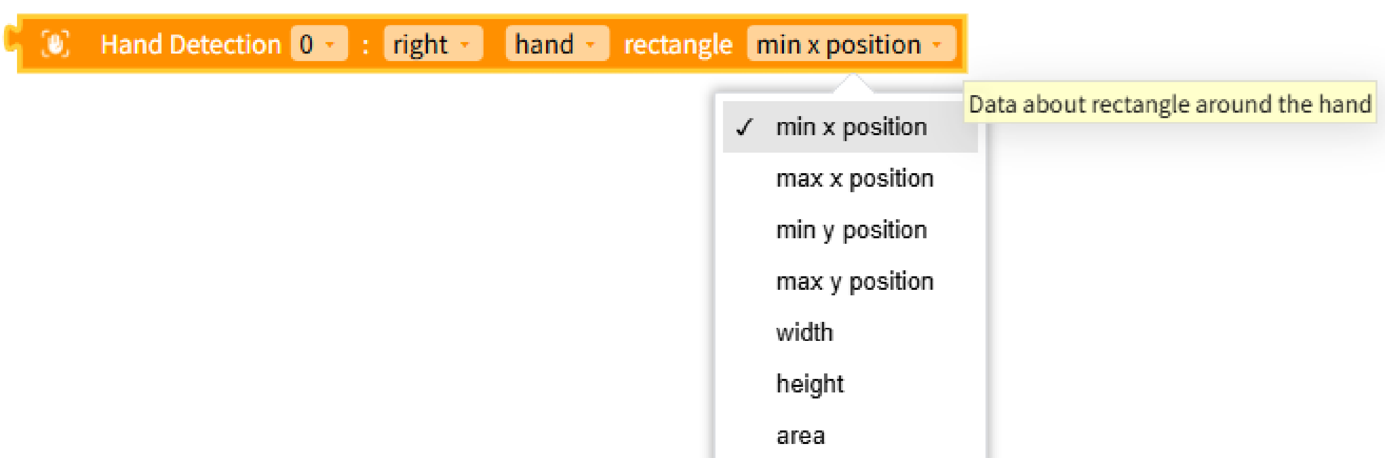
```
$('HandDetection*0:right.pinky.first.y').d;
```

Python Code

```python
# Right thumb first joint x coordinate
__('HandDetection*0:right.thumb.first.x').d

# Right index second joint x coordinate
__('HandDetection*0:right.index.second.x').d

# Right middle third joint x coordinate
__('HandDetection*0:right.middle.third.x').d

# Right ring first joint y coordinate
__('HandDetection*0:right.ring.first.y').d

# Right pinky first joint y coordinate
__('HandDetection*0:right.pinky.first.y').d
```

Hand Bounding Box Data

Defines the area around the detected hand as a rectangle and returns that rectangle's data.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction	Dropdown Option	Hand side	left, right
part	Dropdown Option	Hand part	hand, palm
axis	Dropdown Option	Rectangle data	min_x, max_x, min_y, max_y, width, height, area

JavaScript Code

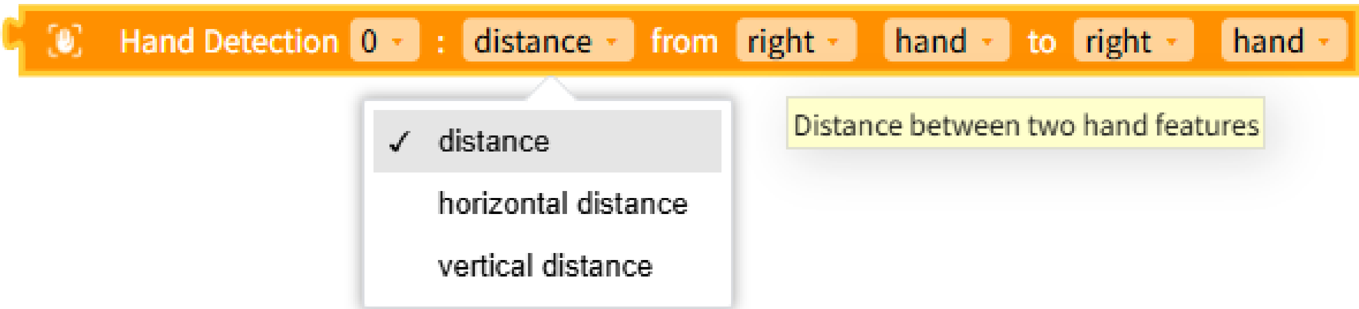
```javascript
// Right hand min x
$('HandDetection*0:right.hand.min_x').d;

// Right hand max x
$('HandDetection*0:right.hand.max_x').d;

// Right hand min y
$('HandDetection*0:right.hand.min_y').d;

// Right hand max y
$('HandDetection*0:right.hand.max_y').d;

// Right hand width
$('HandDetection*0:right.hand.width').d;

// Right hand height
$('HandDetection*0:right.hand.height').d;

// Right hand area
$('HandDetection*0:right.hand.area').d;

// Left palm area
$('HandDetection*0:left.palm.area').d;
```

Python Code

```
# Right hand min x
__('HandDetection*0:right.hand.min_x').d

# Right hand max x
__('HandDetection*0:right.hand.max_x').d

# Right hand min y
__('HandDetection*0:right.hand.min_y').d

# Right hand max y
__('HandDetection*0:right.hand.max_y').d

# Right hand width
__('HandDetection*0:right.hand.width').d

# Right hand height
__('HandDetection*0:right.hand.height').d

# Right hand area
__('HandDetection*0:right.hand.area').d

# Left palm area
__('HandDetection*0:left.palm.area').d
```

Distance Between Two Hands

Returns the distance between two hands.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction 1	Dropdown Option	Hand side	right, left
part 1	Dropdown Option	Hand part	hand, palm
direction 2	Dropdown Option	Hand side	right, left
part 2	Dropdown Option	Hand part	hand, palm
distance	Dropdown Option	Distance type	distance, horizontal distance, vertical distance

JavaScript Code

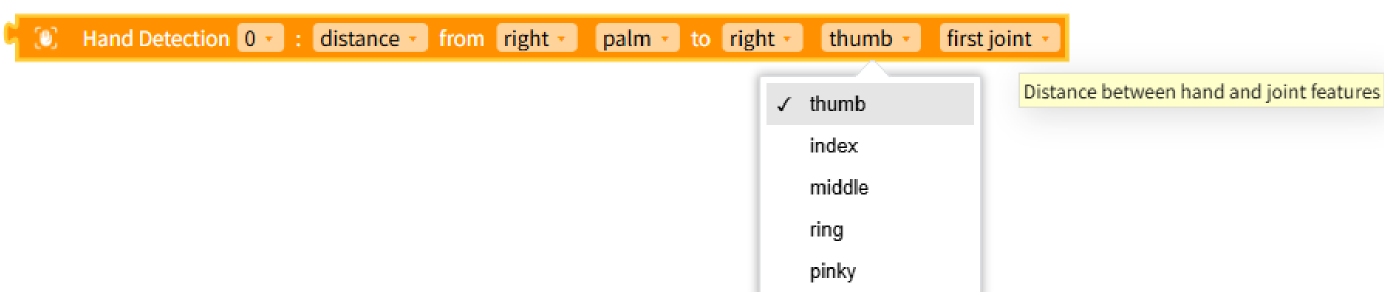
```
// Distance from right hand to left hand
Math.sqrt( Math.pow(($('HandDetection*0:left.hand.x').d -
$('HandDetection*0:right.hand.x').d), 2) +
Math.pow(($('HandDetection*0:left.hand.y').d -
$('HandDetection*0:right.hand.y').d), 2) );

// Horizontal distance from right palm to left palm
Math.abs($('HandDetection*0:left.palm.x').d -
$('HandDetection*0:right.palm.x').d);

// Vertical distance from right hand to left palm
Math.abs($('HandDetection*0:left.palm.y').d -
$('HandDetection*0:right.hand.y').d);
```

Python Code

```
# Distance from right hand to left hand
math.sqrt( math.pow((__('HandDetection*0:left.hand.x').d -
__('HandDetection*0:right.hand.x').d), 2) +
math.pow((__('HandDetection*0:left.hand.y').d -
__('HandDetection*0:right.hand.y').d), 2) )

# Horizontal distance from right palm to left palm
math.fabs(__('HandDetection*0:left.palm.x').d -
__('HandDetection*0:right.palm.x').d)

# Vertical distance from right hand to left palm
math.fabs(__('HandDetection*0:left.palm.y').d -
__('HandDetection*0:right.hand.y').d)
```

Distance Between Hand and Finger Joint

Returns the distance between a hand part and a finger joint.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction 1	Dropdown Option	Hand side	right, left
part 1	Dropdown Option	Hand part	hand, palm
direction 2	Dropdown Option	Hand side	right, left
finger 2	Dropdown Option	Finger	thumb, index, middle, ring, pinky
joint 2	Dropdown Option	Finger joint	first, second, third, last

Name	Type	Description	Range / Type
distance	Dropdown Option	Distance type	distance, horizontal distance, vertical distance

JavaScript Code

```
// Distance from right palm to right thumb first joint
Math.sqrt( Math.pow(($('HandDetection*0:right.thumb.first.x').d -
$('HandDetection*0:right.palm.x').d), 2) +
Math.pow(($('HandDetection*0:right.thumb.first.y').d -
$('HandDetection*0:right.palm.y').d), 2) );

// Horizontal distance from right palm to right index second joint
Math.abs($('HandDetection*0:right.index.second.x').d -
$('HandDetection*0:right.palm.x').d);

// Vertical distance from right palm to right middle third joint
Math.abs($('HandDetection*0:right.middle.third.y').d -
$('HandDetection*0:right.palm.y').d);

// Distance from right palm to right ring last joint
Math.sqrt( Math.pow(($('HandDetection*0:right.ring.last.x').d -
$('HandDetection*0:right.palm.x').d), 2) +
Math.pow(($('HandDetection*0:right.ring.last.y').d -
$('HandDetection*0:right.palm.y').d), 2) );

// Distance from right palm to right pinky last joint
Math.sqrt( Math.pow(($('HandDetection*0:right.pinky.last.x').d -
$('HandDetection*0:right.palm.x').d), 2) +
Math.pow(($('HandDetection*0:right.pinky.last.y').d -
$('HandDetection*0:right.palm.y').d), 2) );
```

Python Code

```
# Distance from right palm to right thumb first joint
math.sqrt( math.pow((__('HandDetection*0:right.thumb.first.x').d -
__('HandDetection*0:right.palm.x').d), 2) +
math.pow((__('HandDetection*0:right.thumb.first.y').d -
__('HandDetection*0:right.palm.y').d), 2) )

# Horizontal distance from right palm to right index second joint
math.fabs(__('HandDetection*0:right.index.second.x').d -
__('HandDetection*0:right.palm.x').d)

# Vertical distance from right palm to right middle third joint
math.fabs(__('HandDetection*0:right.middle.third.y').d -
__('HandDetection*0:right.palm.y').d)

# Distance from right palm to right ring last joint
math.sqrt( math.pow((__('HandDetection*0:right.ring.last.x').d -
__('HandDetection*0:right.palm.x').d), 2) +
math.pow((__('HandDetection*0:right.ring.last.y').d -
__('HandDetection*0:right.palm.y').d), 2) )

# Distance from right palm to right pinky last joint
math.sqrt( math.pow((__('HandDetection*0:right.pinky.last.x').d -
__('HandDetection*0:right.palm.x').d), 2) +
math.pow((__('HandDetection*0:right.pinky.last.y').d -
__('HandDetection*0:right.palm.y').d), 2) )
```

Distance Between Two Finger Joints

Returns the distance between two finger joints.

Dropdown Options and Input Values

Name	Type	Description	Range / Type
direction 1	Dropdown Option	Hand side	right, left
finger 1	Dropdown Option	Finger	thumb, index, middle, ring, pinky
joint 1	Dropdown Option	Finger joint	first, second, third, last
direction 2	Dropdown Option	Hand side	right, left
finger 2	Dropdown Option	Finger	thumb, index, middle, ring, pinky
joint 2	Dropdown Option	Finger joint	first, second, third, last
distance	Dropdown Option	Distance type	distance, horizontal distance, vertical distance

JavaScript Code

```
// Distance from right thumb first joint to left thumb first joint
Math.sqrt( Math.pow(($('HandDetection*0:left.thumb.first.x').d -
$('HandDetection*0:right.thumb.first.x').d), 2) +
Math.pow(($('HandDetection*0:left.thumb.first.y').d -
$('HandDetection*0:right.thumb.first.y').d), 2) );

// Distance from right thumb first joint to left index second joint
Math.sqrt( Math.pow(($('HandDetection*0:left.index.second.x').d -
$('HandDetection*0:right.thumb.first.x').d), 2) +
Math.pow(($('HandDetection*0:left.index.second.y').d -
$('HandDetection*0:right.thumb.first.y').d), 2) );

// Distance from right thumb first joint to left middle third joint
Math.sqrt( Math.pow(($('HandDetection*0:left.middle.third.x').d -
$('HandDetection*0:right.thumb.first.x').d), 2) +
Math.pow(($('HandDetection*0:left.middle.third.y').d -
$('HandDetection*0:right.thumb.first.y').d), 2) );
```

```
// Horizontal distance from right thumb first joint to left ring second joint
Math.abs($('HandDetection*0:left.ring.second.x').d -
$('HandDetection*0:right.thumb.first.x').d);

// Vertical distance from right thumb first joint to left pinky second joint
Math.abs($('HandDetection*0:left.pinky.second.y').d -
$('HandDetection*0:right.thumb.first.y').d);
```

Python Code

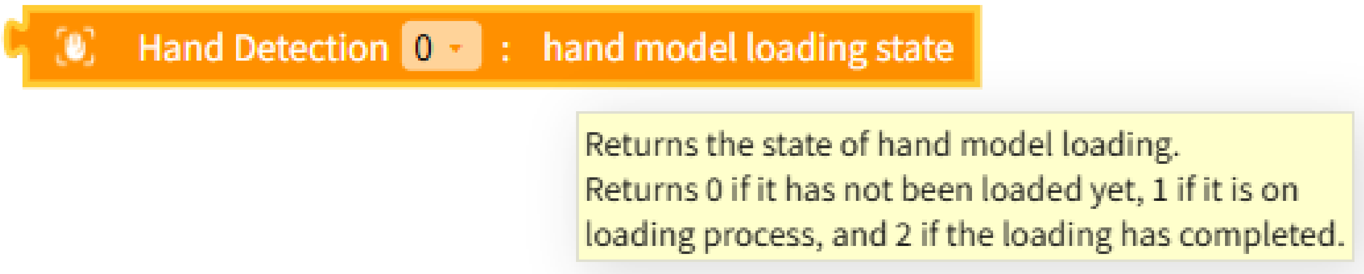
```
# Distance from right thumb first joint to left thumb first joint
math.sqrt( math.pow((__('HandDetection*0:left.thumb.first.x').d -
__('HandDetection*0:right.thumb.first.x').d), 2) +
math.pow((__('HandDetection*0:left.thumb.first.y').d -
__('HandDetection*0:right.thumb.first.y').d), 2) )

# Distance from right thumb first joint to left index second joint
math.sqrt( math.pow((__('HandDetection*0:left.index.second.x').d -
__('HandDetection*0:right.thumb.first.x').d), 2) +
math.pow((__('HandDetection*0:left.index.second.y').d -
__('HandDetection*0:right.thumb.first.y').d), 2) )

# Distance from right thumb first joint to left middle third joint
math.sqrt( math.pow((__('HandDetection*0:left.middle.third.x').d -
__('HandDetection*0:right.thumb.first.x').d), 2) +
math.pow((__('HandDetection*0:left.middle.third.y').d -
__('HandDetection*0:right.thumb.first.y').d), 2) )

# Horizontal distance from right thumb first joint to left ring second joint
math.fabs(__('HandDetection*0:left.ring.second.x').d -
__('HandDetection*0:right.thumb.first.x').d)

# Vertical distance from right thumb first joint to left pinky second joint
```

```
math.fabs(__('HandDetection*0:left.pinky.second.y').d -
__('HandDetection*0:right.thumb.first.y').d)
```

Hand Model State Value

Returns the hand model loading state.

Returns 0 if not loaded, 1 if loading, and 2 if loading is complete.



JavaScript Code

```
// Hand model state value
$('HandDetection*0:model_state').d;
```

Python Code

```
# Hand model state value
__('HandDetection*0:model_state').d
```

Was a Hand Detected?

Returns whether a hand was detected as **True (1) / False (0)**.

JavaScript Code

```
// Was a hand detected?
$('HandDetection*0:detected').d;
```

Python Code

```
# Was a hand detected?
__('HandDetection*0:detected').d
```